

Breaking the Pair: Evaluating Dyadic Interaction via Speaker Switching

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Abstract—Speakers in dialogue continuously adapt their communicative behavior across acoustic, lexical, and semantic dimensions, a phenomenon known as conversational entrainment. Modeling this process requires representations that capture the global structure of interaction, yet prior approaches fail to disentangle dyad-specific patterns from speaker-specific traits, limiting their ability to capture true conversational adaptation. We address this with the Dyadic Distance Matrix (DDM), which encodes all pairwise similarities between the turns of two speakers over an entire conversation, capturing long-range cross-speaker dependencies. This raises a key question: does the DDM represent genuine interaction, or merely reflect individual speaker characteristics? We propose the speaker-switch test, a principled control in which one speaker’s turns are replaced with those from an unrelated speaker drawn from a different conversation. This preserves turn-level statistics while disrupting the original dyadic co-adaptation. The ability to distinguish real from switched DDMs thus directly evaluates whether the representation encodes interaction-specific structure. Across four embedding types and classifiers including ResNet-50 on the CANDOR corpus, real DDMs are consistently distinguishable from their switched counterparts. Comparisons with LibriSpeech show higher discriminability in read speech, highlighting the role of prosodic variability in naturalistic conversations. GradCAM analysis further reveals distinct structural signatures driving classification. These results establish the speaker-switch test as a robust diagnostic for validating representations of dyadic conversational interaction.

Index Terms—conversational entrainment, dyadic distance matrix, speaker-switch, GradCAM, spoken dialogue

I. INTRODUCTION

Speakers in dialogue continuously adapt their communicative behavior to their interlocutors across acoustic, lexical, and semantic dimensions [1]–[3]. This phenomenon, known as *conversational entrainment*, is formalized under Communication Accommodation Theory (CAT) [4], [5], which predicts both convergence and divergence depending on communicative goals. Capturing entrainment computationally therefore requires representations that move beyond local, adjacent-turn similarity to model the evolving global structure of an entire conversation. However, most prior computational approaches rely on local turn-level metrics [6], [7], which conflate two distinct sources of structure: (i) *conversation-specific* patterns arising from the joint behavior of a dyad, and (ii) *speaker-specific* characteristics intrinsic to each participant.

As a result, models may rely on speaker identity rather than true dyadic co-adaptation, limiting their validity as measures of conversational entrainment.

To address this limitation, we use the Dyadic Distance Matrix (DDM), a representation that encodes all pairwise cosine distances between the turn embeddings of two speakers across an entire conversation. By capturing long-range cross-speaker dependencies and global temporal structure, the DDM provides a compact representation of interaction dynamics. This naturally raises a fundamental question: does the DDM encode genuine *interaction*, or does it primarily reflect the statistical properties of individual speakers?

We answer this question through the *speaker-switch test* (Fig. 1), a principled control designed to disrupt dyadic interaction while preserving speaker-level properties. Given a conversation between speakers *A* and *B*, we construct a synthetic counterpart by replacing *B*’s turn sequence with turns from an unrelated speaker drawn from a different conversation. This produces a controlled contrast (e.g., Candor-Real vs. Candor-Switch) that maintains individual feature distributions and turn structure, while eliminating genuine co-adaptation between the original pair. A model that reliably distinguishes real from switched DDMs therefore provides direct evidence that the representation encodes interaction-specific structure. Our contributions are as follows:

- 1) We introduce the speaker-switch test, a principled evaluation framework for distinguishing interaction-specific structure from speaker-specific effects in conversational representations.
- 2) We conduct a systematic evaluation across four embedding types spanning acoustic, structural, and semantic layers, comparing ResNet-50, CNN, and MLP classifiers.
- 3) We perform a cross-corpus analysis using LibriSpeech read speech dataset [8] to examine how prosodic variability in naturalistic speech affects acoustic discriminability.
- 4) We provide a GradCAM-based [9] interpretability analysis identifying which structural regions of the DDM carry discriminative signal.

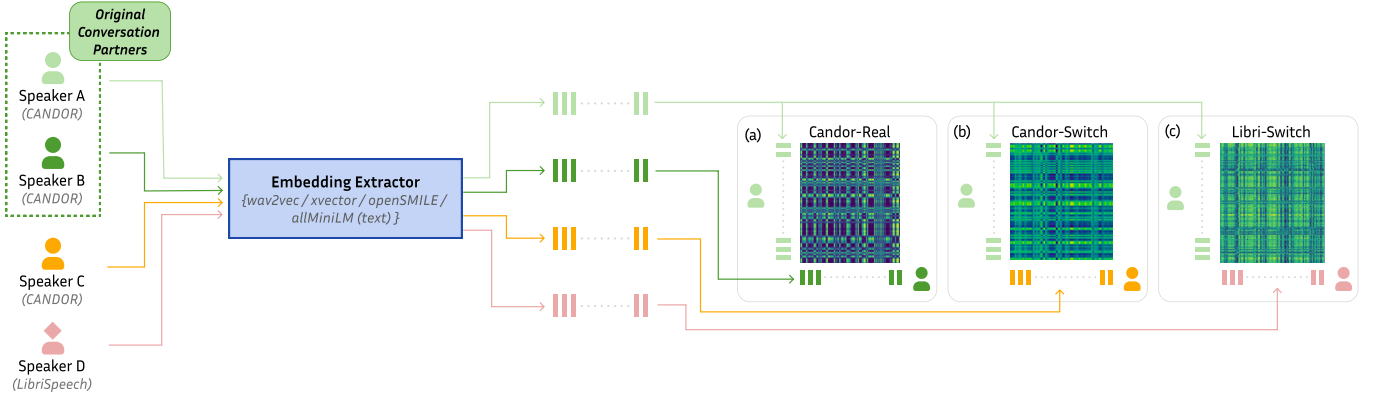


Fig. 1. Dyadic Distance Matrix (DDM) generation pipeline. Turn-level embeddings (a_i, b_j) are extracted using acoustic or semantic models to compute the global cross-speaker distance matrix (M) . The schematic illustrates the formulation of the three experimental pairings: (a) genuine interaction between original conversation partners (Candor-Real), (b) within-corpus control using an unrelated interlocutor (Candor-Switch), and (c) the cross-corpus read-speech baseline (Libri-Switch) using Speaker-D’s speech from LibriSpeech dataset [8].

II. BACKGROUND

A. Dyadic Distance Matrix

We represent a two-speaker conversation using the Dyadic Distance Matrix (DDM), which captures cross-speaker similarity across all turns. Let $\mathbf{a}_i$ and $\mathbf{b}_j$ denote the embedding sequences corresponding to the turns of speakers A and B , respectively. The DDM is defined as a matrix $M \in \mathbb{R}^{|\mathbf{a}| \times |\mathbf{b}|}$, where each entry (i, j) encodes the cosine distance between turns i and j :

$$M_{i,j} = 1 - \frac{\mathbf{a}_i^\top \mathbf{b}_j}{\|\mathbf{a}_i\|_2 \|\mathbf{b}_j\|_2}. \quad (1)$$

Each entry of M reflects the degree of similarity between a pair of turns across speakers, providing a global view of interaction over the entire conversation. Unlike adjacent-turn metrics [6], [7], which focus on local dependencies, the DDM encodes all cross-speaker turn combinations, enabling the analysis of long-range interactional structure.

Following Communication Accommodation Theory (CAT), we compute DDMs across three complementary representational layers: (i) an *affective layer* (wav2vec 2.0, openSMILE), capturing prosodic and paralinguistic cues; (ii) a *structural layer* (x-vector), capturing speaker-specific characteristics and residual identity signals; and (iii) a *semantic layer* (all-MiniLM), capturing lexical and topical alignment between speakers.

III. METHODOLOGY

A. Datasets

We use the CANDOR corpus [10], comprising over 1,600 hours of spontaneous dyadic conversations from more than 1,300 speakers recorded via video-calling platforms. Each session lasts approximately 25 minutes and includes word-level timestamps, speaker diarization, and transcripts. We apply Cliffhanger sentence-level segmentation to obtain semantically coherent speaker turns, and exclude sessions with fewer than 20 turns per speaker.

To assess the role of prosodic variability, we conduct a complementary cross-corpus experiment using LibriSpeech [8]. As a large-scale read-speech corpus, LibriSpeech provides a controlled setting with relatively uniform prosody, in contrast to the highly variable naturalistic speech in CANDOR.

B. Task Formulation

We cast the speaker-switch test as a binary classification task over DDMs. Given a Dyadic Distance Matrix M , the objective is to predict whether it is derived from a *real* conversation (Candor-Real) or a *speaker-switched* counterpart (Candor-Switch).

Formally, the classifier learns a mapping $f : M \rightarrow \{0, 1\}$, where the label indicates the presence or absence of genuine dyadic interaction. Successful discrimination implies that the DDM encodes interaction-specific structure beyond speaker-level statistics.

C. Speaker-Switch DDM Construction

Given an original conversation $C_1 = (A_1, B_1)$ and an unrelated conversation $C_2 = (A_2, B_2)$, the switched DDM for C_1 is constructed by replacing B_1 with B_2 , while retaining A_1 ’s turn sequence. Conversation pairs are matched by approximate turn count to preserve matrix dimensions.

$$M_{i,j}^{\text{sw}} = 1 - \frac{\mathbf{a}_i^{(1)\top} \mathbf{b}_j^{(2)}}{\|\mathbf{a}_i^{(1)}\|_2 \|\mathbf{b}_j^{(2)}\|_2}. \quad (2)$$

This construction preserves: (i) each speaker’s embedding distribution; (ii) within-conversation temporal structure; and (iii) global pairwise distance statistics, while eliminating (iv) genuine dyadic co-adaptation between A_1 and B_1 .

D. Feature Extraction

We evaluate four embedding types corresponding to the three CAT representational layers:

- **wav2vec 2.0** [11]: 768-dimensional representations obtained via mean-pooling of the final transformer layer;

the base pre-trained model is used to retain prosodic information.

- **x-vector** [12]: 512-dimensional speaker embeddings extracted from a pre-trained ECAPA-TDNN model via `pyannote.audio` [13].
- **openSMILE (GeMAPS)** [14], [15]: 88-dimensional acoustic descriptors capturing pitch, energy, spectral, and cepstral features.
- **all-MiniLM** [16]: 384-dimensional sentence embeddings encoding contextual semantic information.

Per-speaker z -normalization is applied to wav2vec 2.0, x-vector, and openSMILE embeddings to remove static speaker-level biases prior to DDM computation.

E. Classifiers

We evaluate three classifier architectures. The primary model is a ResNet-50 backbone [17] implemented in PyTorch [18], with the final fully connected layer replaced by a dropout layer ($p = 0.3$) and a sigmoid output. A three-layer CNN and a multilayer perceptron (MLP) serve as baselines.

All models are trained using the Adam optimizer with learning rate 1×10^{-4} , batch size 32, and early stopping on validation loss. The dataset is split 70/15/15 (train/validation/test) stratified by conversation. DDMs are resized to 64×64 and standardized to unit variance. Evaluation metrics include accuracy, macro-F1, and Equal Error Rate (EER) [19], [20].

F. Cross-Corpus Evaluation

To isolate the effect of prosodic variability, we construct cross-corpus switched DDMs by replacing one speaker in a CANDOR conversation with a randomly sampled speaker from LibriSpeech. Specifically, given a CANDOR conversation $C_1 = (A_1, B_1)$, we retain A_1 's turn sequence and replace B_1 with speech segments from an unrelated LibriSpeech speaker, matched by approximate turn count.

This results in a synthetic interaction between naturalistic conversational speech and read speech, preserving the structure of one side of the dialogue while introducing a controlled mismatch in speaking style. Applying the same classification pipeline allows us to assess how reduced prosodic variability in read speech affects discriminability relative to fully naturalistic dyadic interactions.

G. GradCAM Interpretability

We apply Gradient-weighted Class Activation Mapping (GradCAM) [9] to the trained ResNet-50 to identify discriminative regions within the DDM. For each correctly classified test sample, GradCAM heatmaps are computed and averaged across samples within each condition and embedding type. These aggregated maps reveal the structural loci of interaction-specific information captured by the model.

IV. RESULTS AND DISCUSSION

A. Classification: Real vs. Switch

Table I presents classification results (CANDOR Switch) across all modalities, model architectures, and corpora.

On CANDOR, the semantic embedding (all-MiniLM) achieves perfect discrimination with ResNet-50 (Acc=1.000, EER=0.000), and remains highly discriminable even with the shallow CNN (Acc=0.952) and MLP (Acc=0.857). This confirms that conversation-specific topical and lexical alignment is the strongest signal of genuine dyadic interaction, robust to classifier choice.

For acoustic and structural embeddings on CANDOR, ResNet-50 consistently outperforms the shallower architectures. x-vector ResNet-50 achieves Acc=0.682 versus CNN at 0.597, while the MLP collapses to chance (Acc=0.500, F1=0.000), indicating it cannot learn the interaction patterns encoded in the structural DDM. A similar collapse occurs for openSMILE MLP (Acc=0.467, below chance), whereas ResNet-50 attains 0.692. This architecture sensitivity confirms that acoustic interaction signatures in the DDM are complex and spatially distributed, requiring hierarchical feature extraction to detect.

That all four embedding types exceed chance on CANDOR with at least one model provides validation: DDMs encode genuine interaction-specific structure that cannot be recovered by combining arbitrary speakers. Semantic representations are most discriminable because lexical and topical co-adaptation is intrinsically tied to the specific dyad; a switched conversation cannot share the same topical trajectory.

B. Cross-Corpus Results: The Role of Prosodic Variability

The contrast between CANDOR and LibriSpeech results is the most striking finding in Table I. For x-vector embeddings, all three classifiers achieve perfect accuracy on LibriSpeech (Acc=1.000, EER=0.000), including the MLP that collapses on CANDOR. LibriSpeech consists of read speech where intonation is controlled and within-speaker variability is low. In this setting, x-vector embeddings robustly encode stable, distinct speaker signatures, making cross-speaker pairing trivially detectable regardless of classifier capacity.

On CANDOR, speakers exhibit rich intonation variation, voice modulation, and prosodic accommodation. Post- z -normalization, these residual patterns of within-speaker variability and cross-speaker adaptation reduce the discriminability of switched DDMs, and only ResNet-50 retains meaningful performance. The acoustic embedding results similarly follow this pattern: wav2vec 2.0 and openSMILE both show markedly higher accuracy on LibriSpeech than on CANDOR across all models. The LibriSpeech result thus serves as an important sanity check: the difficulty of distinguishing real from switched acoustic DDMs on CANDOR is not a failure of the representation, but rather a reflection of the genuine complexity of conversational acoustic adaptation. This finding aligns with the cross-database turn-filling approach, where analogous inferences were drawn about the role of vocal variability in turn-level influence.

Notably, the MLP achieves Acc=1.000 on wav2vec 2.0 LibriSpeech despite underperforming ResNet-50 on CANDOR. This inversion suggests that read speech embeddings contain a simple low-dimensional discriminative feature exploitable by

TABLE I
REAL VS. SWITCH CLASSIFICATION RESULTS ACROSS MODALITIES, MODELS, AND CORPORA. BOLD DENOTES THE BEST ACCURACY PER MODALITY AND CORPUS.

Modality	Model	CANDOR Switch			LibriSpeech Switch		
		Acc	F1	EER	Acc	F1	EER
wav2vec 2.0	ResNet-50	0.620	0.649	0.394	0.904	0.904	0.092
	CNN	0.552	0.433	0.474	0.823	0.820	0.169
	MLP	0.584	0.559	0.402	1.000	1.000	0.000
x-vector	ResNet-50	0.682	0.687	0.315	1.000	1.000	0.000
	CNN	0.597	0.572	0.388	1.000	1.000	0.000
	MLP	0.500	0.000	0.485	1.000	1.000	0.000
openSMILE	ResNet-50	0.692	0.722	0.350	0.908	0.904	0.083
	CNN	0.575	0.557	0.450	0.842	0.832	0.150
	MLP	0.467	0.458	0.533	0.975	0.974	0.000
all-MiniLM	ResNet-50	1.000	1.000	0.000	0.998	0.998	0.004
	CNN	0.952	0.950	0.064	0.994	0.994	0.004
	MLP	0.857	0.858	0.149	1.000	1.000	0.000

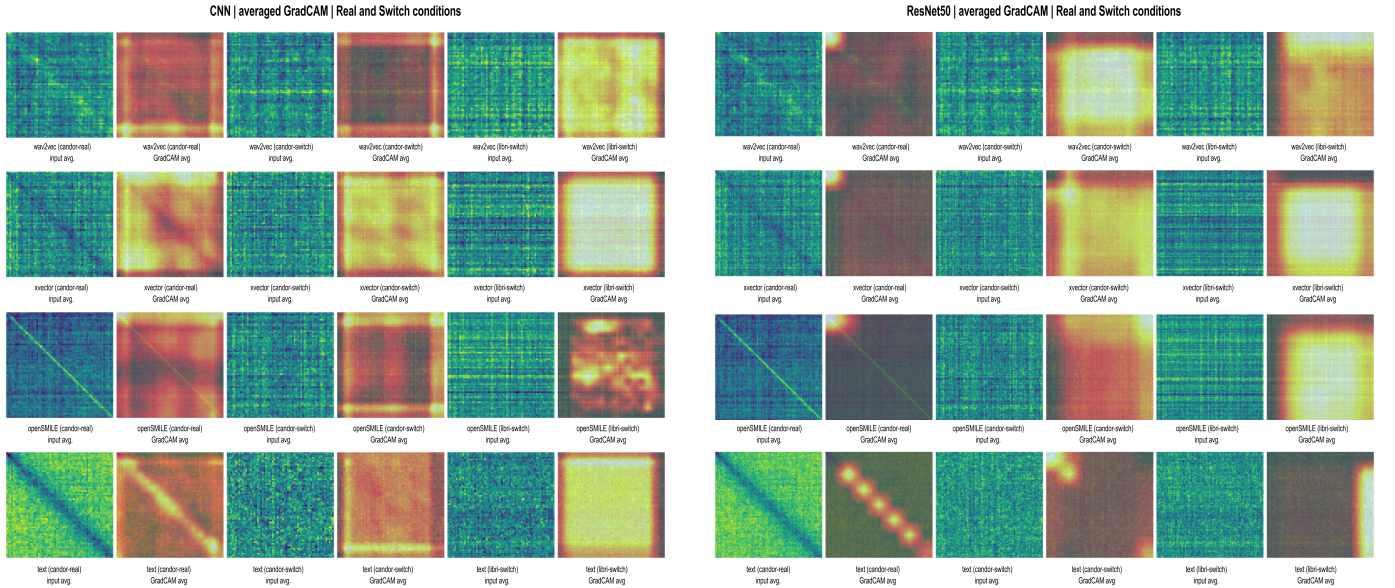


Fig. 2. Mean GradCAM heatmaps for ResNet-50 and CNN models across all embedding types (columns: wav2vec 2.0, x-vector, openSMILE, all-MiniLM). Rows show averaged input DDMs and GradCAM activations for real and switch conditions, where warmer colors indicate higher activation.

a linear-style classifier, while conversational speech requires multi-scale spatial reasoning to separate real from switched patterns.

C. GradCAM Analysis

Fig. 2 shows mean GradCAM activations for ResNet-50 and CNN, aggregated over correctly classified test samples under real and switch conditions.

Semantic modality: For all-MiniLM, ResNet-50 GradCAM maps under the real condition exhibit strong activation concentrated along and near the DDM diagonal. This diagonal structure reflects temporal proximity: turns occurring close in time are semantically similar, consistent with topic progression

and shared reference in natural conversation. Under the switch condition, the diagonal activation disappears, replaced by diffuse or off-diagonal patterns, confirming that the diagonal is a signature of genuine conversational content alignment rather than any individual speaker property.

Acoustic modality: For wav2vec 2.0 and x-vector, ResNet-50 activates more globally rather than concentrating on the diagonal. This is consistent with the literature documenting acoustic accommodation operating over longer time scales [1]: prosodic adaptation is distributed across the full conversation timeline rather than localized to adjacent turns. The x-vector switch condition shows notably different global activation patterns, indicating the classifier detects the absence of structural

coherence in the switched DDM.

ResNet-50 vs. CNN: Comparing Figs. 2 and 2, the CNN exhibits broader and less spatially structured activations, particularly for acoustic modalities. This explains the performance gap on CANDOR: the hierarchical feature extraction of ResNet-50 enables detection of complex, spatially distributed interaction signatures, whereas the three-layer CNN lacks sufficient depth.

The MLP collapse on acoustic CANDOR DDMs shows these patterns are not linearly separable, highlighting that modeling natural conversational accommodation requires deep, non-linear architectures capable of capturing distributed dyadic behavior.

V. CONCLUSION

We have presented a systematic speaker-switch validation of Dyadic Distance Matrices as representations of conversational interaction. Across four embedding types on CANDOR, switched DDMs are consistently distinguishable from real DDMs, validating that the DDM encodes genuine dyad-specific interaction structure beyond individual speaker characteristics. The LibriSpeech cross-corpus experiment reveals prosodic adaptation is an attribute of conversational speech. GradCAM analysis shows the structural signatures in the DDMs that drive classification: diagonal temporal alignment for semantic features and distributed global structure for acoustic features. These findings provide an interpretable foundation for using DDMs as a representation of conversational dynamics. Future work will integrate these validated DDM representations into predictive architectures. By isolating true co-adaptation from speaker-specific traits, DDMs may enable tasks like outcome prediction, mutual understanding assessment, and more responsive dialogue systems.

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